DESIGN AND FABRICATION OF AN AUTOMATED SMART COMPOST BIN WITH BLENDING AND COMPOST POWDER CONDITIONING SYSTEM

PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the course

of

GEA1401 – ENVIRONMENTAL SCIENCE (TCPR)

IN

DEPARTMENT OF

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM,TRICHY

ANNAUNIVERSITY

CHENNAI 600 025

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Under the Guidance of

Dr.M.CHRISPIN DAS M.E., Ph.D

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BONAFIDE CERTIFICATE

Certified that this project report titled **"DESIGN AND FABRICATION OF AN AUTOMATED SMART COMPOST BIN WITH BLENDING AND COMPOST POWDER CONDITIONING SYSTEM"** is the Bonafide work of **RATHEESHSARMA J (8115U23ME036), SHARANRAJ C(8115U23ME043), YAMUTHAN P (8115U23ME063), AVINASH K (8115U23ME301),** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Under
ANNA UNIVERSITY

CHENNAI DECLARATION BY THE CANDIDATES

We jointly declare that the project report on **"DESIGN AND FABRICATION OF AN AUTOMATED SMART COMPOST BIN WITH BLENDING AND COMPOST POWDER CONDITIONING SYSTEM"** is the result of original work done by us and best of our knowledge, similar work has not been submitted to "ANNA UNIVERSITY CHENNAI" for the requirement of Degree of BACHELOR OF ENGINEERING. This project report is submitted on the partial fulfillment of the requirement of the award of degree of BACHELOR OF ENGINEERING.

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DEPARTMENT OF MECHANICAL ENGINEERING
Program: Bachelor of Engineering

VISION & MISSION OF THE COLLEGE

VISION:

- To achieve a prominent position among the top technical institutions

MISSION:

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To provide education that can be used as a tool to transform the students.

VISION & MISSION OF THE DEPARTMENT

VISION:

- To provide eminent Mechanical Engineers to the society and industries through quality technical education and research.

MISSION:

- To impart quality professional education through state-of-the-art-infrastructure and resources.
- To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.
- To carryout collaborative projects with industries and academics with fullest potential.

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Program: Bachelor of Engineering

Program Educational Objectives :(PEOs)

Graduates of Mechanical Engineering shall:

1. Have Successful careers in industries or opt for higher studies and research.

2. Analyze, synthesize and design mechanical and allied products addressing societal issues.

3. Exhibit technical expertise, communication skills, team work with professionalism with ethical values.

Program Specific Outcomes (PSOs)

Students shall:

PSO1: Identify, formulate and solve engineering problems in three core streams of Mechanical Engineering, i.e., Design, Thermal and Manufacturing Engineering

PSO2: Design, develop, test and implement energy efficient systems for required Engineering applications.

PROGRAM OUTCOMES

PROGRAM OUTCOMES (PO's)	
PO 1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledg and research methods including design of experiments, analysis an interpretation of data, and synthesis of the information to provide vali conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

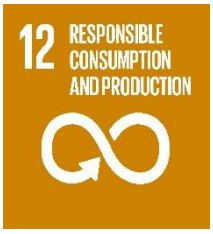
PROGRAM's OUTCOME MAPPING

Check	PO number	Explanation
☑	**PO1 – Engineering Knowledge**	Applied heat transfer, airflow, and mechanical design principles to develop the grain dryer.
☑	**PO2 – Problem Analysis**	Identified grain-drying challenges and considered ways to reduce moisture and post-harvest losses.
☑	**PO3 – Design/Development of Solutions**	Designed a dryer to meet grain-processing needs while considering safety and usability.
☑	**PO4 – Conduct Investigations**	Tested the dryer and evaluated temperature, drying time, and grain condition.
☑	**PO5 – Modern Tool Usage**	Used suitable design, fabrication, and monitoring tools during project development.
☑	**PO6 – Engineer and Society**	Considered farmers' needs and the safe use of grain-drying equipment.
☑	**PO7 – Environment and Sustainability**	Aimed to reduce grain wastage and support sustainable use of harvested produce.
☑	**PO8 – Ethics**	Followed responsible engineering practices and reported project results honestly.
☑	**PO9 – Individual and Team Work**	Contributed to the team's design, fabrication, and project activities.
☑	**PO10 – Communication**	Presented the design, working method, and results through reports and presentations.
☑	**PO11 – Project Management and Finance**	Planned project tasks, selected components, and considered the project budget.
☑	**PO12 – Life-long Learning**	Learned about grain-drying technology and considered opportunities for future improvement.

SUSTAINABLE DEVELOPMENT GOAL



PRIMARY SDG's USED IN THE PROJECT



SDG 12: Responsible Consumption and Production: SDG 12 focuses on reducing waste through prevention, recycling, and reuse. The proposed compost bin processes biodegradable organic waste using mechanical blending and a heating plate to reduce excess moisture. It promotes responsible waste management and the productive use of organic waste.

SECONDARY SDG's USED IN THE PROJECT



SDG 11: Sustainable Cities and Communities: SDG 11 aims to make communities more sustainable. The proposed compost bin helps manage biodegradable waste through mechanical blending and compost conditioning, supporting better waste-management practices in households, hostels, and educational institutions.



SDG 12: Responsible Consumption and Production: The dryer aims to make better use of harvested produce by reducing losses linked to poor moisture management. Its enclosed chamber can limit exposure to external contaminants, while controlled operation can help reduce the risk of insufficient or excessive drying. The goal is to preserve grain quality and support more responsible handling after harvest.

ABSTRACT

Organic waste management is an important aspect of maintaining a clean and sustainable environment. Improper disposal of biodegradable waste can cause unpleasant odours, sanitation problems, and environmental pollution. Traditional composting methods often require considerable time and space, making them less convenient for households, hostels, and educational institutions.This project proposes the design and fabrication of a compost bin with an integrated mechanical blending and heating system. The proposed system consists of a composting chamber, a mechanical blending mechanism, a motor, a heating plate, and a supporting structure. The blending mechanism helps mix the organic waste, while the heating plate assists in reducing excess moisture from the material. These features are intended to improve waste handling and support the preparation of compostable organic material.The system aims to provide a convenient and practical approach to processing biodegradable waste. The mechanical blending process promotes uniform mixing, while controlled heating can help reduce moisture content. Proper temperature control and suitable operating conditions are important to avoid excessive heating that may affect beneficial microorganisms involved in composting.The project aims to encourage responsible waste management, reduce the amount of organic waste sent for disposal, and promote the productive use of biodegradable materials. Experimental testing is required to evaluate blending performance, moisture reduction, energy consumption, processing time, and the quality of the final compost.

Keywords: *Compost bin, organic waste management, mechanical blending, heating plate, moisture reduction, biodegradable waste, sustainable waste management.*

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LIST OF ABBREVATION

S.NO	ABBREVATION	WORD
1	SUSTAINABLE DEVELOPMENT GOAL	SDG
2	DIRECT CURRENT	DC
3	WET BASIC	WB
4	DRY BASIC	DB
5	STAINLESS STEEL	SS
6	COMPUTER AIDED DRAWING	CAD
7	ALTERNATIVE CURRENT	AC

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CHAPTER-1
INTRODUCTION

1.1 GENERAL INTRODUCTION

Organic waste management has become an important concern due to the increasing generation of biodegradable waste from households, hostels, educational institutions, and agricultural activities. A significant portion of this waste consists of vegetable peels, fruit residues, leftover food, and other organic materials. Improper disposal of such waste can lead to unpleasant odours, unhygienic surroundings, and environmental pollution. Therefore, effective methods are required to convert biodegradable waste into useful products while reducing the amount of waste sent for disposal.

Composting is a biological process in which microorganisms decompose biodegradable organic materials into nutrient-rich compost under suitable conditions. The resulting compost can be used as an organic soil amendment to improve soil fertility and support plant growth. However, conventional composting methods often require manual mixing, regular handling, and sufficient time for decomposition. Uneven mixing and inadequate aeration can also affect the decomposition process. These limitations highlight the need for a practical system that simplifies compost handling and supports efficient processing.

The Design and Fabrication of an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System aims to address these challenges through a mechanically driven blending arrangement and a dedicated compost powder conditioning unit. The system consists of a mild-steel structural frame, a vertical composting chamber, a geared motor, a central shaft with blending blades, a conical hopper, and a heating plate with a tray. The geared motor drives the central shaft to mix the organic material, while the heating plate and tray provide a means of conditioning the processed compost by reducing excess moisture.

The proposed system integrates blending and compost powder conditioning within a compact structure to reduce manual handling and improve the convenience of compost processing. The project focuses on developing a cost-effective and practical solution for biodegradable waste management, with potential applications in households, hostels, and educational institutions. By converting organic waste into useful compost material, the system supports sustainable waste utilization and environmentally responsible practices.

The project is intended to provide a practical approach to biodegradable waste processing in locations such as households, hostels, educational institutions, and other small-scale waste-generation environments. By incorporating mechanical blending and a dedicated conditioning section, the apparatus aims to reduce manual effort during material handling and support the preparation of processed compost for subsequent use. The proposed system is designed as a cost-conscious prototype that can be fabricated using commonly available materials and mechanical components.

Overall, the project demonstrates the application of mechanical design and fabrication principles to organic waste management. It brings together structural fabrication, geared-motor operation, shaft-and-blade mixing, gravity-assisted material movement, and heating-based compost conditioning in a single apparatus. The development of this system supports the broader objective of converting biodegradable waste into a useful resource while encouraging improved waste-handling practices. Its performance can be assessed through prototype testing based on blending operation, material handling, moisture reduction, energy consumption, and the quality of the final conditioned compost.

1.2 FEATURES

The proposed automated smart compost bin incorporates several mechanical features to support organic waste processing and compost powder conditioning. The major features of the system are described below. The vertical composting chamber provides a compact arrangement for holding biodegradable organic waste. A geared motor drives the central shaft and attached blending blades to facilitate mechanical mixing of the organic material. The conical hopper guides the processed material towards the lower section of the apparatus.

The blending mechanism consists of a geared motor, a central shaft, and a set of blending blades. The geared motor provides the rotational motion required to operate the shaft and blades inside the composting chamber. The rotating blades facilitate the mixing of organic materials and help distribute the waste more uniformly throughout the chamber. This mechanical arrangement is intended to simplify the mixing operation and reduce the manual effort required during compost processing.

A conical hopper is provided at the lower portion of the composting chamber to guide the processed material towards the subsequent section. Its tapered shape supports the

downward movement of material under gravity and provides a convenient transition between the main chamber and the lower conditioning arrangement. The use of a vertically arranged chamber and conical hopper helps maintain a compact layout while integrating the major mechanical components.

The compost powder conditioning unit consists of a heating plate and a tray. The heating plate supplies thermal energy to the processed compost material, assisting in the reduction of excess moisture. The tray holds the material during the conditioning operation and facilitates its collection after processing. This arrangement is intended to improve the handling characteristics and consistency of the compost, depending on the heating conditions and duration of treatment.

The mild-steel structural frame provides support for the composting chamber, motor assembly, hopper, and conditioning unit. Mild steel is suitable for prototype fabrication because it is readily available and can be shaped and assembled using conventional workshop processes. The overall design emphasizes structural simplicity, ease of fabrication, and the use of commonly available mechanical components.

The integration of the blending mechanism and compost powder conditioning unit provides a practical arrangement for small-scale organic waste processing. The system aims to reduce manual handling during mixing and conditioning while supporting the conversion of biodegradable waste into useful compost material. Its compact construction and cost-conscious approach make it potentially suitable for applications in households, hostels, and educational institutions.

Another important feature of the proposed system is the use of a geared motor for controlled mechanical operation. The reduction gearing provides a suitable rotational speed for the blending mechanism, allowing the central shaft and blades to mix the organic material at a relatively low speed. This arrangement is intended to provide stable operation while accommodating the mechanical requirements of the prototype. The motor is mounted in alignment with the central shaft to establish a direct mechanical drive for the blending assembly.

CHAPTER-2

LITERATURE REVIEW

[1] Manuetal. (2019): Investigated decentralized composting of household wet biodegradable waste using plastic drums.The findings emphasized the importance of regular turning and suitable aeration for effective organic waste decomposition.**Outcome**: Demonstrated the suitability of low-cost drum composting systems for household organic waste management.

[2] Kalamdhad and Kazmi (2009): Investigated the composting of different organic waste mixtures in a rotary drum composter. Their results showed that the composition of the waste mixture strongly influenced compost maturity and quality. **Outcome**: Established the importance of waste composition and mechanical mixing in rotary drum.

[3] Guo et al. (2012) Designed and tested an in-store paddy-drying system, considering airflow distribution, grain quality, energy requirements, and economic performance.The study also demonstrated uniform airflow through the tested arrangement and recommended periodic ventilation during storage.**Outcome:** Highlighted the importance of blower selection and air distribution for achieving consistent drying.

[4] Alkoaik (2019): Investigated the integration of continuous aeration and rotation during the composting of agricultural residues.stationary reactor under aerated conditions to evaluate the influence of rotation on the active composting phase.**Outcome:** Showed that combining rotation with aeration can influence the composting process and reduce the duration.

[5] Bhave and Kulkarni (2019): Examined active and passive aeration methods for decentralized composting of household biodegradable waste.Active aeration reduced the maturation period compared with passive aeration in the experimental setup.**Outcome:** Demonstrated that the aeration method can influence compost maturation time.

[6] Rao and Parsai (2023): Presented a comprehensive review of decentralized composting systems for household biodegradable waste management.the maturity of compost.**Outcome**: Highlighted decentralized composting as a practical approach to household organic waste management.

[7] Lai et al. (2024): Investigated aeration management strategies for small-scale food waste composting.temperature development, moisture loss, and compost maturity.**Outcome:** Demonstrated that appropriate aeration arrangements can improve the operating condition.

[8] Zhu et al. (2024): Studied alternating ventilation during food waste composting and its influence on temperature, oxygen, and moisture distribution within a static composting reactor. The research examined how ventilation management affected mineralization and humification**.Outcome:**Highlighted the importance of managing airflow and internal composting conditions to support organic matter decomposition

[9] Schneider et al. (2020): Investigated radial mixing performance in a model drum composting system.rotating drum affected the distribution of the composting substrate.**Outcome:** Provided useful engineering insights into material movement and mixing behaviour in drum-based composting.

[10] Aliiev et al. (2022): Examined the mechanized process of organic waste composting. The research considered mechanical arrangements and process requirements associated with handling and mixing organic materials during compost production.**Outcome**: Emphasized the role of mechanization in improving the handling and processing of organic waste.

[11] Saypariya et al. (2024): Studied the composting of the organic fraction of municipal solid waste using a three-stage biodegradable composter. The research evaluated the treatment of organic waste through successive processing stages and considered the quality of the resulting compost.**Outcome:** Demonstrated the potential of staged composting arrangements for processing the organic fraction of municipal solid waste.

[12] Colak and Hepbasli (2009), Part 1: Reviewed heat-pump drying systems, their historical development, operating arrangements, and mathematical models.It provides supporting knowledge for comparing conventional heated-air equipment with more complex air-conditioning arrangements. **Outcome:** Established a framework for understanding how heat-pump system configuration and operating conditions influence drying performance.

[13] Jolanun et al. (2005): Investigated the influence of moisture content on a fed-batch composting reactor treating vegetable and fruit waste. The research considered how moisture conditions affected the composting process and the operation of the reactor.**Outcome:** Emphasized the importance of maintaining suitable moisture conditions.

[14] Rasapoor et al. (2016): Compared different aeration methods for municipal solid waste composting.The study assessed practical aeration approaches and their influence on the composting process.**Outcome:** Showed that aeration strategy is an important consideration when selecting and operating composting systems.

[15] Singh et al. (2024): Assessed the feasibility of composting sewage sludge in a rotary drum reactor.mechanically operated reactor for treating organic sludge and evaluated its suitability for composting applications.**Outcome:** Provided evidence supporting the investigation of rotary drum reactors for controlled organic waste.

[16] Alkoaik et al. (2018): Investigated the energy performance of a rotary drum bioreactor used for composting tomato plant residues operating the reactor and processing agricultural waste.**Outcome:** Highlighted the importance of evaluating energy consumption.

[17] Kalamdhad and Kazmi (2009): Studied the effects of turning frequency on compost stability. The research examined how the frequency of material turning influenced the composting process.**Outcome:** Identified turning frequency as an important operating factor.

[18] Kalamdhad and Kazmi (2009): Investigated the influence of the carbon-to-nitrogen ratio on mixed organic waste composting in a rotary drum composter. The study examined how variations in the initial waste mixture influenced the composting process and the resulting material.**Outcome:** Emphasized the importance of selecting a suitable waste mixture to obtain stable and mature compost.

[19] Shivam et al. (2025): Investigated the bioconversion of food waste into compost using a rotary drum composter for agricultural utilization. The study explored the application of drum-based composting to process food waste and produce material for potential agricultural use.**Outcome:** Supported the use of rotary drum composting as an approach for converting food waste into a potentially useful soil amendment.

[20] Das and Dhal (2026): Presented a review of rotary drum composting for efficient organic waste management. The review discussed the application of rotary drum composters, their operating principles, and their potential role in processing organic waste.**Outcome:** Summarized the relevance of rotary drum composting as a mechanical approach to organic.

CHAPTER-3

BACKGROUND STUDY

3.1 OVERVIEW OF ORGANIC WASTE COMPOSTING

Organic waste management is an important part of sustainable environmental practices. Biodegradable materials such as vegetable peels, fruit waste, leftover food, and other organic residues are generated daily in households, hostels, hotels, and educational institutions. When these materials are disposed of without proper treatment, they can contribute to unpleasant odours, sanitation problems, and environmental pollution. Composting provides a practical method of managing such waste by converting biodegradable materials into a useful organic product.

Composting is a biological process in which microorganisms break down organic matter under suitable environmental conditions. During this process, factors such as moisture content, temperature, oxygen availability, and the composition of the waste influence decomposition and the quality of the final compost. Proper handling and mixing of organic materials can help distribute moisture and organic matter more uniformly throughout the composting material.

Conventional composting methods often involve manual turning and handling, which require time and physical effort. In small-scale applications, these operations may become inconvenient when biodegradable waste is generated regularly. Mechanical composting systems can help simplify material mixing by using a motor-driven shaft and blending blades to move and distribute the organic material within a chamber.

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3.2 PRINCIPLE OF COMPOSTING

Composting is a biological process in which microorganisms decompose biodegradable organic materials into a stable, humus-like product. The process depends on several factors, including moisture content, temperature, oxygen availability, and the composition of the organic waste. Suitable conditions support microbial activity and help convert waste materials into useful compost.

During composting, microorganisms break down complex organic substances into simpler compounds. This activity generates heat and gradually changes the physical and chemical characteristics of the waste. The progress of decomposition depends on the type of material used and the environmental conditions maintained during the process. Proper moisture levels are important because excessive dryness can limit microbial activity, while excessive moisture can restrict oxygen movement within the material.

3.3 MECHANICAL COMPONENTS AND THEIR FUNCTIONS

The proposed compost bin consists of several mechanical components designed to support the blending and conditioning of biodegradable organic waste. The major components include a mild-steel structural frame, a vertical composting chamber, a geared motor, a central mixing shaft, blending blades, a conical hopper, a heating plate, and a tray. Each component performs a specific function and contributes to the overall operation of the system.

The mild-steel structural frame provides support for the composting chamber, motor assembly, hopper, and conditioning section. Mild steel is selected because of its availability, strength, and suitability for conventional fabrication processes such as cutting, welding, and drilling. The frame maintains the arrangement of the major components and provides structural stability during operation.

3.4 WORKING PRINCIPLE OF THE PROPOSED SYSTEM

The proposed system combines mechanical blending and thermal conditioning to support the processing of biodegradable organic waste. The major operations are carried out using a vertical composting chamber, a motor-driven mixing mechanism, a conical hopper, and a heating plate with a tray. These components are arranged within a mild-steel structural frame to provide a compact and practical assembly.

Initially, the biodegradable organic material is placed inside the composting chamber. The geared motor supplies rotational motion to the central mixing shaft, which is connected to the blending blades. As the shaft rotates, the blades move through the material and facilitate mixing within the chamber. This arrangement reduces dependence on manual mixing and helps distribute the material more uniformly.

3.5 MATERIAL SELECTION AND FABRICATION METHODS

Material selection is an important consideration in the design and fabrication of mechanical equipment. The materials selected for the proposed compost bin must provide adequate structural support, allow convenient fabrication, and remain suitable for the intended operating conditions. The selection also considers availability, cost, ease of assembly, and the practical requirements of the prototype.

Mild steel is used for the structural frame and major supporting components of the proposed system. It is commonly used in mechanical fabrication because it offers suitable strength, is readily available, and can be cut, drilled, and welded using conventional workshop equipment. Its use also helps keep the overall fabrication cost within the intended budget.

3.6 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed Design and Fabrication of an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System offers several practical advantages for small-scale organic waste processing. The system combines a motor-driven blending mechanism with a separate heating plate and tray, providing an integrated arrangement for material mixing and conditioning.

One of the main advantages is the reduction in manual effort required for mixing. The geared motor drives the central shaft and blending blades, allowing the organic material to be mixed mechanically inside the composting chamber. This arrangement reduces the need for repeated manual turning and simplifies the mixing operation.The central mixing shaft and blending blades were prepared and assembled inside the chamber. The geared motor was securely mounted and mechanically connected to the shaft to transmit rotational motion for blending the organic material.

CHAPTER-4

PROBLEM IDENTIFICATION

The generation of biodegradable organic waste is a common issue in households, hostels, hotels, and educational institutions. Materials such as vegetable peels, fruit residues, and leftover food require proper handling and disposal. When these materials are not managed effectively, they can create unpleasant odours, sanitation problems, and difficulties in waste handling. Composting provides a method for converting biodegradable waste into useful organic material; however, conventional small-scale composting practices may involve considerable manual effort.

4.1 IDENTIFICATION OF THE PROBLEM

Traditional sun drying relies on sunlight, natural airflow, and favourable atmospheric conditions. During rainy or cloudy weather, the available heat may be insufficient to dry harvested grains within the required time. High atmospheric humidity can also slow moisture removal, extending the drying period.

Unexpected rainfall requires farmers to collect or cover the grain quickly. Repeated spreading and collection increase labour and interrupt the drying process. If partially dried grain becomes wet again, further drying is necessary before storage. These difficulties create a need for an enclosed drying system that reduces dependence on direct sunlight and provides more consistent operating conditions.

4.2 NEED FOR THE PROPOSED SYSTEM

The identified problems indicate the need for a compact and economical system that simplifies the mechanical mixing and conditioning of biodegradable organic material. The proposed project addresses these requirements through a vertical composting chamber, a geared motor, a central mixing shaft, blending blades, a conical hopper, and a heating plate with a tray.

The motor-driven blending mechanism is intended to reduce the manual effort associated with mixing. The conical hopper supports downward material movement, while the heating plate and tray provide a separate arrangement for heat-assisted moisture.

4.3 OBJECTIVES OF THE PROPOSED SYSTEM

The main objective of this project is to design and fabricate an automated smart compost bin with a blending and compost powder conditioning system for small-scale organic waste processing. The system is intended to combine mechanical mixing and heat-assisted conditioning within a compact structure using commonly available materials and components.

The specific objectives of the project are as follows:

1. To design and fabricate a vertical composting chamber supported by a mild-steel structural frame.
2. To develop a motor-driven blending mechanism consisting of a geared motor, central shaft, and blending blades to facilitate the mixing of organic material.
3. To incorporate a conical hopper that guides processed material towards the lower conditioning section through gravity-assisted movement.
4. To provide a heating plate and tray for heat-assisted conditioning of the processed compost material and reduction of excess moisture under suitable operating conditions.
5. To minimize manual effort during the blending and material-handling operations.

4.4 SCOPE OF THE PROJECT

The scope of this project is to design and fabricate a compact mechanical system for blending biodegradable organic material and conditioning the processed compost. The proposed system integrates a vertical composting chamber, a geared motor, a central mixing shaft, blending blades, a conical hopper, and a heating plate with a tray. These components are arranged within a mild-steel structural frame to provide a practical and economical prototype.

The project focuses on the mechanical blending of organic material using a motor-driven shaft and blending blades. The rotation of the blades facilitates mixing within the chamber and aims to reduce the manual effort required for this operation.

4.5 LIMITATIONS OF THE PROPOSED SYSTEM

The proposed Design and Fabrication of an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System is developed as a small-scale prototype.

Although the system integrates mechanical blending and heat-assisted conditioning, certain limitations must be considered during its design, operation, and evaluation.

The blending mechanism depends on the geared motor, central shaft, and blending blades to mix the organic material. Its effectiveness may vary depending on the quantity, moisture content, composition, and physical characteristics of the material. Excessively wet, compact, or fibrous waste may affect the movement of the blades and increase the mechanical load during operation.

4.6 EXPECTED OUTCOMES OF THE PROJECT

The proposed Design and Fabrication of an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System is expected to demonstrate the practical integration of mechanical blending and heat-assisted conditioning for small-scale organic waste processing. The system combines a vertical composting chamber, geared motor, central mixing shaft, blending blades, conical hopper, heating plate, and tray within a mild-steel structural frame.

The motor-driven blending mechanism is expected to facilitate the mixing of biodegradable organic material inside the composting chamber. By automating the rotational movement of the blending blades, the system aims to reduce the manual effort required for mixing and improve the convenience of material handling. The effectiveness of blending will depend on the characteristics and quantity of the material processed.

4.7 PROBLEM STATEMENT

The improper management of biodegradable organic waste generated in households, hostels, and educational institutions creates difficulties in waste handling and disposal. Conventional composting practices often involve manual mixing and repeated handling of organic materials, requiring additional time and physical effort. Furthermore, transferring processed material between different stages and managing excess moisture can create operational difficulties in small-scale compost processing.

CHAPTER-5
IMPACT OF THE PROBLEM

This chapter discusses the major impacts associated with improper biodegradable waste management, including environmental pollution, manual effort, and difficulties in handling excess moisture. It also explains the importance of adopting practical and economical waste-processing methods. The proposed compost bin aims to address these challenges through mechanical blending and heat-assisted conditioning, supporting more convenient and organized processing of organic waste.

5.1 ENVIRONMENTAL IMPACT

Improper management of biodegradable organic waste can create environmental and sanitation problems. Organic materials such as vegetable peels, fruit residues, and leftover food may produce unpleasant odours and attract insects when they are not handled appropriately. When biodegradable waste is disposed of in landfills, its decomposition under anaerobic conditions can also contribute to greenhouse gas emissions. These issues highlight the importance of adopting suitable methods for managing organic waste.

Composting provides an approach for converting biodegradable materials into useful organic matter. However, the effectiveness of composting depends on appropriate operating conditions and proper handling of the material. The proposed compost bin aims to support this process through mechanical blending and a separate conditioning arrangement, encouraging more organized handling of biodegradable waste.

5.2 IMPACT OF MANUAL PROCESSING

Conventional small-scale composting often requires manual turning, mixing, and transferring of organic material. These activities can demand considerable time and physical effort, especially when waste is generated regularly. Inadequate or irregular mixing may also result in uneven distribution of material within the composting container.

The proposed system uses a geared motor connected to a central shaft and blending blades to facilitate mechanical mixing. This arrangement is intended to reduce the manual effort

associated with the blending operation. The conical hopper also provides a gravity-assisted path for moving processed material towards the lower conditioning section.

5.3 IMPACT OF EXCESS MOISTURE

Excess moisture can affect the handling, storage, and physical characteristics of processed compost. Material with high moisture content may become sticky, difficult to transfer, and less convenient to store. Therefore, moisture management is an important consideration during compost processing and conditioning.

The proposed system incorporates a heating plate and tray to assist in reducing excess moisture from the processed material. The extent of moisture reduction depends on the initial moisture content, heating temperature, treatment duration, and quantity of material. Appropriate operating conditions are necessary to achieve consistent conditioning without causing undesirable changes to the material.

5.4 ECONOMIC AND OPERATIONAL IMPACT

The cost and complexity of waste-processing equipment can limit its use in households, hostels, and educational institutions. Systems that require specialized equipment or extensive manual handling may be less practical for small-scale applications. A compact design using readily available materials and conventional fabrication methods can help address these practical limitations.

The proposed compost bin uses a mild-steel structural frame, a geared motor, a central mixing shaft, blending blades, a conical hopper, and a heating plate with a tray. Integrating these components into one assembly provides a practical arrangement for blending and conditioning organic material. The design emphasizes economical fabrication, straightforward mechanical operation, and convenient material handling.

5.5 OVERALL IMPACT OF THE PROPOSED SYSTEM

The identified problems demonstrate the need for practical methods to improve the handling and processing of biodegradable organic waste. Manual mixing, material transfer, and excess moisture can create operational difficulties in small-scale compost processing. Addressing these issues can support more organized waste management and improve the convenience of handling processed organic material.

The proposed system aims to address these challenges through motor-driven blending, gravity-assisted material movement, and heat-assisted conditioning. Its effectiveness must be assessed through prototype testing, including the operation of the blending mechanism, movement of material, structural stability, and moisture-conditioning performance.

Overall, the project applies mechanical engineering principles to a practical waste-management problem. Although the prototype does not independently guarantee compost maturity or a specific environmental benefit, it demonstrates an integrated mechanical approach to blending biodegradable material and conditioning the resulting compost.

From an economic and practical perspective, the use of mild steel and commonly available mechanical components supports the fabrication of a cost-conscious prototype. Conventional workshop operations such as cutting, welding, drilling, and assembly can be used to construct the system. This approach also provides practical experience in mechanical design, material selection, component integration, and prototype development.

The proposed system may be useful in small-scale settings such as households, hostels, and educational institutions where biodegradable waste is generated regularly. Its suitability for these applications depends on the quantity and type of waste processed, the operating capacity of the motor, and the performance of the conditioning section. Regular inspection and maintenance of the motor, shaft, blending blades, and heating arrangement are also important for safe and reliable operation.

The project additionally demonstrates how mechanical engineering principles can be applied to address a practical environmental challenge. By combining motor-driven blending, gravity-assisted material movement, and heat-assisted conditioning, the system provides an integrated approach to organic material processing. However, the prototype does not independently guarantee complete composting, compost maturity, or a specific reduction in environmental pollution.

CHAPTER-6
OBJECTIVES

The main objective of this project is to design and fabricate an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System for small-scale organic waste processing. The project focuses on integrating a motor-driven blending mechanism with a heat-assisted conditioning section to simplify material handling and reduce manual effort. The system is developed using commonly available materials and conventional fabrication techniques to maintain a practical and economical design.

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6.1 PRIMARY OBJECTIVE

The primary objective of this project is to design and fabricate an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System that provides a practical mechanical approach to small-scale organic waste processing. The system aims to integrate a motor-driven blending mechanism and a separate heat-assisted conditioning section within a compact structure, thereby simplifying the handling and processing of biodegradable organic material.

The proposed system is designed with a vertical composting chamber supported by a mild-steel structural frame. A geared motor is mechanically connected to a central mixing shaft fitted with blending blades. When the motor operates, the shaft rotates and drives the blades through the organic material, facilitating mixing within the chamber. This arrangement is intended to reduce the manual effort required for mixing and provide a convenient method of handling the material during processing.

Another primary objective is to establish a suitable material flow arrangement between the blending chamber and the conditioning section. A conical hopper is positioned below the chamber to guide the processed material downward under gravity. This arrangement supports the transfer of material towards the lower section while maintaining a compact configuration of the major components.

The project also aims to incorporate a heating plate and tray for conditioning the processed material. The heating plate supplies thermal energy to assist in reducing excess

moisture, while the tray holds the material during the conditioning operation. The effectiveness of this process depends on the initial moisture content, heating temperature, treatment duration, and quantity of material. Therefore, the conditioning section is intended to support moisture reduction rather than independently guarantee complete composting or compost maturity.

In addition, the project emphasizes economical fabrication through the use of mild steel, commonly available mechanical components, and conventional workshop processes. The structural arrangement is developed to provide adequate support for the chamber, motor assembly, hopper, and conditioning section. Proper alignment of the motor and shaft, secure mounting of the blending blades, and stable assembly of the heating section are important considerations in achieving reliable operation.

The overall objective is to develop a functional prototype that combines mechanical blending, gravity-assisted material movement, and heat-assisted conditioning in a single assembly. The completed system can be evaluated through prototype testing to assess the operation of the blending mechanism, material movement, structural stability, ease of handling, and moisture-conditioning performance. Through this project, mechanical engineering principles are applied to the practical challenge of improving the handling and processing of biodegradable organic material.

6.2 NEED FOR THE PROJECT

The increasing generation of biodegradable organic waste creates a need for practical methods to handle and process organic materials efficiently. Conventional composting practices often involve manual mixing, repeated handling, and transferring of materials between different stages. These activities can require considerable time and physical effort, particularly in small-scale applications such as households, hostels, and educational institutions.

The proposed project addresses these practical difficulties by integrating a motor-driven blending mechanism with a separate compost powder conditioning section. The geared motor drives a central shaft fitted with blending blades, facilitating the mixing of organic material inside the vertical chamber. This arrangement is intended to reduce manual effort and simplify the blending operation.

The project also aims to establish a suitable monitoring arrangement. Temperature measurements should be taken at the air inlet, within the chamber, and at representative product locations during testing. Relative humidity should be monitored to understand changes in drying-air conditions. Grain moisture must be verified separately using a calibrated moisture meter or an appropriate laboratory method, because air humidity does not directly indicate the moisture content of the grain.

A further objective is to regulate heating according to the selected operating temperature and provide a controlled transition to cooling. The controller should maintain the required conditions within the capability of the selected hardware. Once the drying endpoint is confirmed, heating should stop and the cooling stage should operate until the product reaches an appropriate unloading condition. The operating sequence should be clear and repeatable.

Protection against overheating and inadequate airflow is also a technical objective. The system should include suitable electrical protection and a means of stopping heating when unsafe conditions occur. Heating should depend on adequate blower operation so that excessive heat does not accumulate in the air path or chamber. Guards and secure component mounting should support safe operation and maintenance.

Performance evaluation forms another major objective. Testing should determine drying duration, water removed, final moisture variation, cooling time, and total electricity consumption. Measurements from repeated trials will help establish whether the system operates consistently. Calculating energy consumption per kilogram of water removed will support comparison between different operating conditions.

The final technical objective is to provide a structure that is maintainable, stable, and practical to move. Trays and filters should be removable for cleaning, while essential components should remain accessible for inspection and repair. The frame and wheel arrangement should support the complete operating load without excessive movement or instability. These assessments will help determine whether the fabricated dryer meets both its processing and handling requirements.

CHAPTER-7
HARDWARE REQUIREMENTS

Hardware components play an important role in the design and fabrication of the proposed Automated Smart Compost Bin with Blending and Compost Powder Conditioning System. The system consists of mechanical and electrical components that work together to facilitate the blending of biodegradable organic material and the conditioning of the processed compost.

7.1 MILD-STEEL STRUCTURAL FRAME

Mild steel is used for the structural frame and supporting parts of the proposed system. It provides a suitable foundation for mounting the composting chamber, motor assembly, hopper, and conditioning section. Mild steel is widely used in mechanical fabrication because it is readily available and can be cut, drilled, welded, and assembled using conventional workshop equipment.

The frame is designed to maintain the position of the major components and provide structural stability during operation. Proper fabrication and welding are important to ensure that the assembly can support the weight of the chamber and other installed components.



Fig 7.1.1 - Mild-steel structural frame

7.2 COLLECTION TRAY

The tray holds the processed material during the conditioning operation. It is positioned in association with the heating plate to allow heat transfer to the material. The tray also provides a convenient means of collecting and removing the material after conditioning.

The tray should be sufficiently strong for the intended material load and arranged to allow convenient handling. Its dimensions and position must be compatible with the heating plate and the overall structural layout.

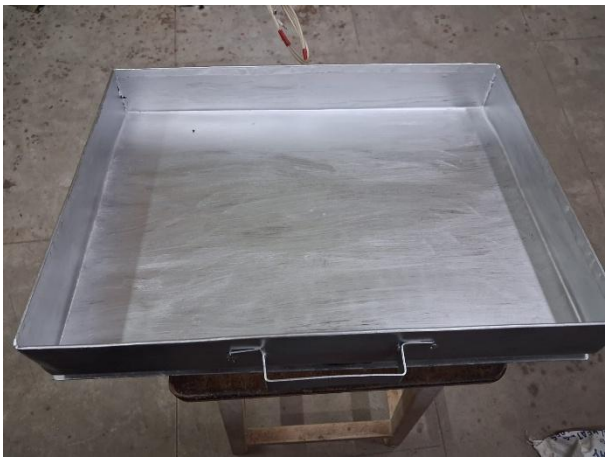


Fig 7.2.1 – Collection tray

7.3 HEATING PLATE

The heating plate supplies thermal energy to the material during the conditioning operation. It forms part of the lower conditioning section and is intended to assist in reducing excess moisture from the processed compost material.

The heating plate must be securely mounted and operated under suitable temperature and time conditions. Since heating performance depends on the material quantity and initial moisture content, the operating conditions should be evaluated during prototype testing. Appropriate electrical connections and precautions against accidental contact with hot surfaces are also necessary.



Fig 7.3.1 – Heating plate

7.4 GEARED MOTOR



Fig 7.4.1 – Geared motor

The geared motor is the main driving component of the blending mechanism. It converts electrical energy into rotational motion and transmits the required movement to the central mixing shaft. The gearbox reduces the motor's rotational speed and provides the output motion needed for the blending operation.

The motor is mounted at the upper portion of the composting chamber and mechanically connected to the central shaft. Correct alignment and secure coupling are important to support

smooth operation. The motor must be suitable for the material load and mechanical requirements of the blending mechanism.

7.5 CONICAL HOPPER

The conical hopper is positioned beneath the composting chamber to guide processed material towards the lower conditioning section. Its tapered shape supports downward material movement under gravity and provides a transition between the main chamber and the tray arrangement.

The hopper must be securely attached to the structure and positioned to facilitate material transfer. Its dimensions and outlet arrangement should be appropriate for the type and quantity of material handled by the prototype. This shape supports gravity-assisted movement of the material and helps direct it towards the collection area without requiring a separate material-conveying mechanism. The hopper is positioned securely within the supporting frame to maintain proper alignment with the chamber and the conditioning section.



Fig 7.5.1- Conical hopper

7.6 BLENDING BLADES

The blending blades are used to support the mixing and circulation of water within the cooling system. They help maintain uniform water distribution inside the cooling tank and prevent temperature differences in different regions. The blade arrangement is designed to promote effective mixing and improve the overall cooling performance of the system.

The blades must be securely attached to the rotating mechanism and positioned properly inside the tank to ensure smooth operation. Their size, shape, and arrangement should be suitable for the tank dimensions and mixing requirements. Proper blending helps maintain uniform temperature distribution and supports efficient heat transfer between the circulating water and the cooling arrangement. This improves the consistency and reliability of the closed-loop cooling process.

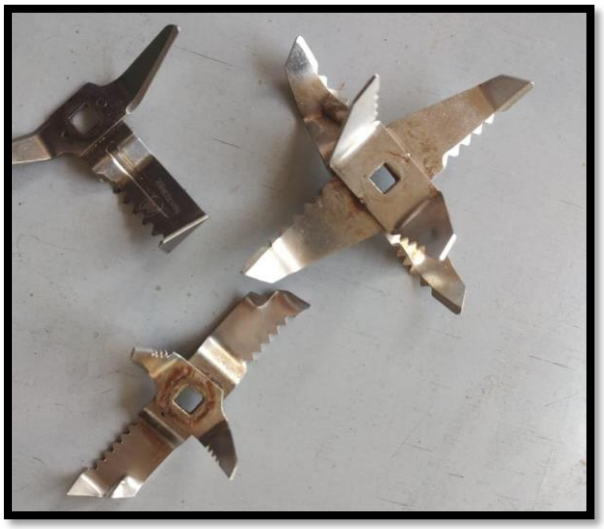


Fig 7.5.1- Conical hopper

CHAPTER-8
WORKING PRINCIPLE

8.1 LOADING OF ORGANIC MATERIAL

The process begins by placing suitable biodegradable organic material inside the vertical composting chamber. The chamber holds the material during the blending operation and provides space for the rotation of the central shaft and blending blades.

The material should be loaded in a quantity appropriate for the dimensions of the chamber and the capability of the motor. Excessive loading may restrict blade movement and increase the mechanical load on the shaft and motor. Proper loading helps facilitate the blending operation and supports convenient handling of the material.

8.2 OPERATION OF THE BLENDING MECHANISM

After loading the material, the geared motor is switched on using the appropriate electrical supply. The motor produces rotational motion, which is transmitted through its output shaft or coupling to the central mixing shaft inside the chamber.

The blending blades attached to the central shaft rotate along with it. Their movement facilitates mixing and redistribution of the organic material within the chamber. This mechanical arrangement is intended to reduce the manual effort required for mixing.

8.3 MOVEMENT OF MATERIAL THROUGH THE HOPPER

Following the blending operation, the processed material is directed towards the lower section through the conical hopper. The tapered shape of the hopper supports downward movement under gravity and provides a transition between the vertical chamber and the conditioning arrangement.

The movement of material depends on the shape of the hopper, the size of its outlet, and the physical characteristics of the processed material. Suitable dimensions and positioning are therefore important to facilitate material transfer and reduce the possibility of blockage.

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8.4 HEAT-ASSISTED CONDITIONING

The lower conditioning section consists of a heating plate and a tray. The processed material is placed on the tray, and the heating plate supplies thermal energy to assist in reducing excess moisture.

During this stage, heat is transferred from the plate to the material in contact with or positioned above it. The extent of moisture reduction depends on the initial moisture content, heating temperature, treatment duration, material quantity, and distribution of the material across the tray.

8.5 COLLECTION OF CONDITIONED MATERIAL

After the conditioning operation is completed, the heating plate is switched off and the material is allowed to cool sufficiently before handling. The conditioned material can then be removed from the tray for further use or storage, depending on its condition and suitability.

The material should be assessed before being used as compost. Mechanical blending and moisture reduction alone do not confirm that the organic waste has completed biological decomposition. Appropriate composting conditions and sufficient decomposition time remain necessary to obtain mature compost.

After the heat-assisted conditioning operation is completed, the heating plate is switched off and the processed material is allowed to cool sufficiently before collection. This step is important because the tray and heating plate may remain hot after the power supply is disconnected. Suitable precautions must be taken while handling the tray and removing the conditioned material to prevent accidental burns.

8.6 OVERALL WORKING SEQUENCE

The overall working sequence of the proposed Automated Smart Compost Bin with Blending and Compost Powder Conditioning System involves a series of operations, beginning with the loading of biodegradable organic material and ending with the collection of the conditioned material. Each stage is designed to perform a specific function within the integrated mechanical assembly.

Initially, suitable biodegradable organic material is loaded into the vertical composting chamber. The geared motor is then operated to transmit rotational motion to the central mixing shaft, which carries the blending blades. The rotating blades facilitate the mixing and redistribution of the organic material within the chamber, helping to reduce the need for repeated manual mixing.

After the blending operation, the processed material is directed towards the lower section through the conical hopper. The tapered configuration of the hopper supports gravity-assisted movement and provides a connection between the blending chamber and the conditioning arrangement. The material is then placed in the tray associated with the heating plate for the conditioning stage.

During conditioning, the heating plate supplies thermal energy to the material to assist in reducing excess moisture. The effectiveness of this stage depends on the initial moisture content, quantity of material, heating temperature, and duration of treatment. Suitable operating conditions must be maintained to prevent excessive heating and undesirable changes to the material.

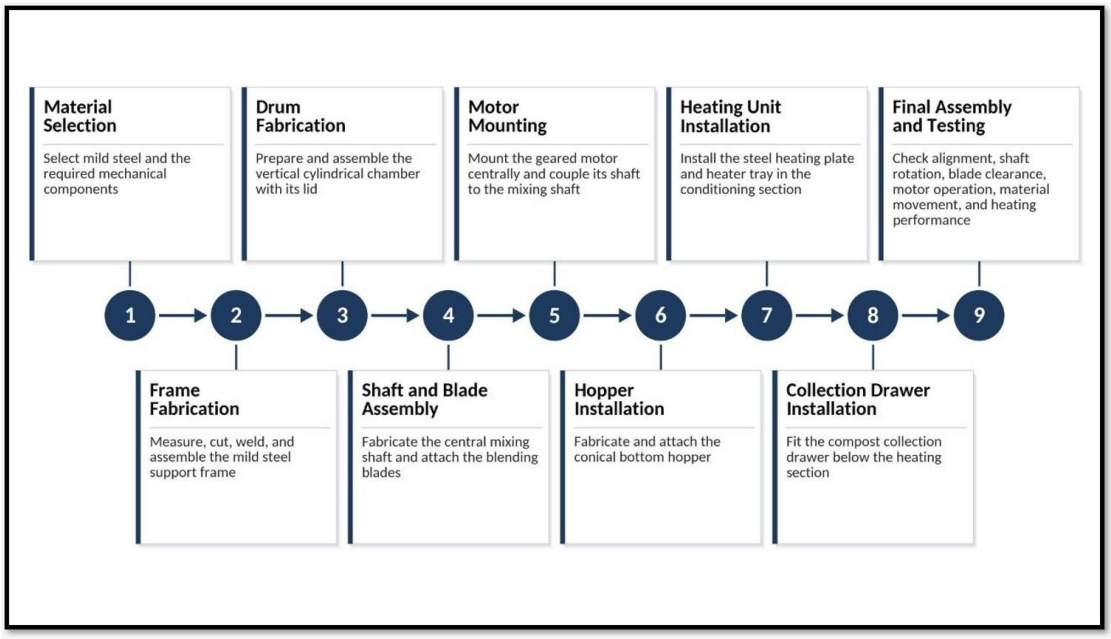


Fig 8.1 Flow diagram

CHAPTER-9

2D & 3D DESIGN

9.1 2D DESIGN

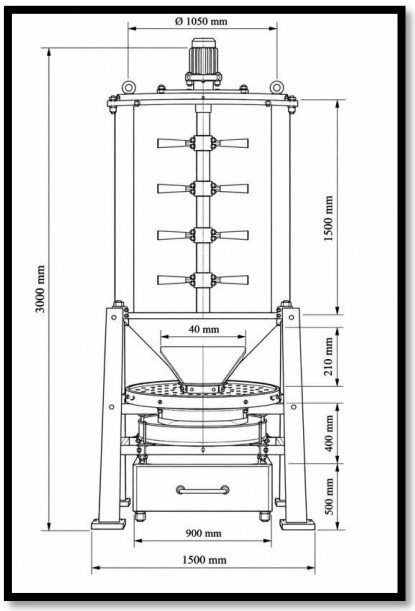


Fig 9.1.1 FRONT VIEW

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The front view illustrates the overall arrangement of the compost bin system, including the composting chamber, geared motor, central mixing shaft, blending blades, conical hopper, heating section, and collection tray. The dimensions shown provide a reference for the height, width, and positioning of the major components, helping explain the structural layout and assembly of the proposed system.

The front view illustrates the overall arrangement of the compost bin system, including the composting chamber, geared motor, central mixing shaft, blending blades, conical hopper, heating section, and collection tray. The dimensions shown provide a reference for the height, width, and positioning of the major components, helping explain the structural layout and assembly of the proposed system.

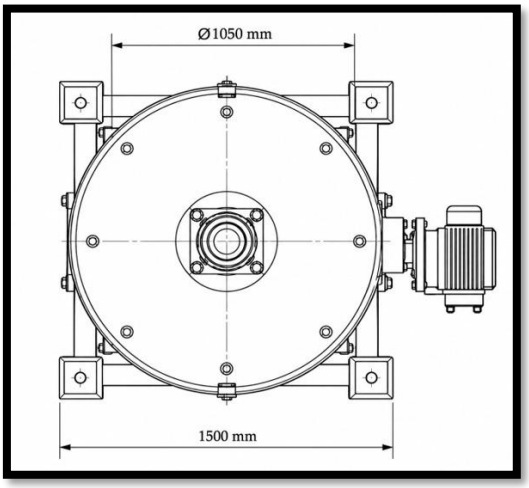


Fig 9.1.2 TOP VIEW

The top view illustrates the circular composting chamber with a diameter of 1050 mm and an overall mounting width of 1500 mm. The central shaft is connected to the mixing mechanism, while the geared motor is positioned on the side to provide rotational motion. The supporting brackets and mounting points ensure structural stability and proper alignment of the system components.

9.2 3D DESIGN



Fig 9.2.1 TOP VIEW

The front view illustrates the overall construction of the composting system, including the cylindrical composting chamber, top-mounted geared motor, central mixing shaft, and blending blades. The conical hopper directs the processed compost material into the lower collection unit. The supporting frame provides structural stability, while the bottom collection tray facilitates the collection of the finished compost for further use.



Fig 9.2.2 FRONT VIEW

The front view shows the top cover of the composting system, with the geared motor mounted at the center to drive the mixing mechanism. The circular lid is secured with multiple locking clamps to prevent unwanted opening during operation. The supporting frame provides stability and supports the main chamber. This arrangement allows easy access for maintenance, inspection, and cleaning of the composting unit.

The front view shows the overall arrangement of the composting machine, including the cylindrical composting chamber, top-mounted geared motor, central mixing shaft, and horizontal mixing blades. The conical hopper directs the processed organic waste into the lower collection unit. The supporting frame provides stability and strength, while the collection tray facilitates the easy removal of the finished compost. This view helps explain the positioning and assembly of the major components of the proposed system.



Fig 9.2.3 ISOMETRIC VIEW

The isometric view illustrates the complete three-dimensional arrangement of the composting system, including the cylindrical composting chamber, top-mounted geared motor, central mixing shaft, and horizontal mixing blades. The conical hopper guides the processed organic waste toward the lower collection unit. The supporting frame provides strength and stability to the system, while the collection tray allows easy removal of the finished compost. This view clearly represents the overall design, component placement, and structural arrangement of the proposed composting machine.

The isometric view provides a three-dimensional representation of the proposed composting machine, showing the overall structure and arrangement of its major components. It consists of a cylindrical composting chamber, a top-mounted geared motor, a central mixing shaft with horizontal blades, and a conical hopper. The supporting frame ensures stability, while the bottom collection unit collects the processed compost.

CHAPTER-10

PROTOTYPE MODEL

10.1 INTRODUCTION

The prototype model of the Automated Smart Compost Bin with Blending and Compost Powder Conditioning System was fabricated by integrating the major mechanical components into a single unit. A 25-litre water can was modified and spray-painted to serve as the main composting chamber. It was mounted vertically within a mild-steel supporting frame to provide stability. A geared motor was fitted at the top of the chamber and connected to the internal mixing shaft to drive the blending blades.

The lower section consists of a conical outlet arrangement that guides the processed material towards the heat-assisted conditioning unit. A sheet-metal enclosure houses the heating arrangement, while a separate collection drawer is provided beneath it to collect the processed material. The complete assembly demonstrates the integration of the blending mechanism, material transfer arrangement, heating plate.

The prototype model of the proposed Automated Smart Compost Bin with Blending and Compost Powder Conditioning System was developed to demonstrate the practical implementation of the designed system. The prototype was fabricated by assembling the major mechanical and electrical components into a compact and functional unit.

The main composting chamber was constructed using a 25-litre water can, which was modified and spray-painted to provide a suitable container for organic waste processing. The chamber was mounted vertically on a mild-steel supporting frame to ensure proper alignment, stability and durability during operation.

A geared DC motor was installed at the upper portion of the composting chamber. The motor was coupled with a central shaft carrying blending blades, which help in mixing and breaking down the organic waste. This arrangement ensures uniform mixing of the compost materials and supports the decomposition process.

The lower portion of the composting chamber was designed with a conical-shaped outlet to facilitate the controlled movement of processed organic material towards the next

stage. This arrangement helps prevent material accumulation inside the chamber and provides a smooth transfer path.

A separate conditioning unit was fabricated using a sheet-metal enclosure. It consists of a heating arrangement designed to reduce the moisture content of the processed compost material. The heating unit helps convert the semi-processed compost into a relatively dry material suitable for further powder preparation.A collection drawer was provided at the bottom of the conditioning unit to collect the final processed compost material. This makes the collection process easier and helps maintain a clean working environment.

The entire prototype was assembled by integrating the composting chamber, geared motor, blending shaft, blades, supporting frame, conical outlet, heating arrangement and collection drawer. The arrangement demonstrates the overall working concept of the proposed system, from organic waste mixing to the collection of conditioned compost material.



Fig. 10.1 Prototype Model of Smart Compost Bin

CHAPTER-11
COST ESTIMATION

11.1 Bill Of Materials

S.No.	Item	Specification / purpose	Quantity
1	Mild steel	Structural frame and supporting components	1 lot
3	Geared motor	Drives the central mixing shaft and blending blades	1
4	Conical hopper	Guides processed material towards the conditioning section	1
5	Central mixing shaft	Transmits rotational motion to the blending blades	1
6	Blending blades	Mixes and redistributes organic material inside the chamber	1 set
7	Heating plate	Supplies heat for moisture conditioning	3
8	Collection tray	Holds material during conditioning and collection	1

Tab 11.1- Bill Of Materials

11.2 TABLE OF ESTIMATION

S.No	Component / work	Estimated cost (₹)
1	Mild steel	250
2	Geared motor	150
3	Conical hopper fabrication	400
4	Central mixing shaft fabrication	350
5	Blending blade fabrication	350
6	Heating plate	2,500
7	Collection tray	500
	Estimated total project cost	₹4500/-

Tab 11.2 – Cost Estimation

CHAPTER-12

CONCLUSION

The project entitled “Design and Fabrication of an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System” was successfully designed and fabricated to provide an efficient method for processing biodegradable organic waste. The prototype consists of a mild-steel frame, vertical chamber, geared motor, mixing shaft, blending blades, conical hopper, heating plate, and collection tray.

In this project, we fabricated the supporting frame, assembled the motor-driven mixing mechanism, and integrated the heating and material collection sections. The blending blades help to mix and break down organic waste, while the conical hopper transfers the material to the conditioning section. The heating plate helps to reduce excess moisture in the processed material.

Through this project, we have successfully integrated the blending and heat-assisted conditioning mechanisms into a single compact prototype. The developed system helps to reduce manual handling, simplify organic waste processing, and improve the convenience of compost preparation. The fabrication process also provided practical experience in material selection, cutting, welding, drilling, machining, motor mounting, shaft alignment, and mechanical assembly.

From this project, we conclude that the developed system provides a simple and practical approach to organic waste processing by combining blending and moisture conditioning in a single unit. It reduces manual effort and makes compost preparation more convenient. Further improvements in automation and temperature control can enhance the overall efficiency and performance of the system.

Overall, the project successfully achieves its objective of developing a compact and economical prototype for sustainable organic waste management.

CHAPTER-13
FUTURE SCOPES

The proposed project, titled “Design and Fabrication of an Automated Smart Compost Bin with Blending and Compost Powder Conditioning System,” provides a foundation for further development in small-scale organic waste processing. The present design integrates a motor-driven blending mechanism with a heating plate and collection tray for compost powder conditioning. Although the prototype focuses on basic mechanical operation and heat-assisted moisture reduction, several improvements can be explored in the future to enhance its performance, safety, and suitability for different applications.

One possible improvement is the optimization of the blending mechanism. The dimensions, arrangement, and shape of the blending blades can be modified to improve the movement and distribution of organic material within the chamber. Different blade configurations and rotational speeds can be tested to determine their influence on mixing performance. The motor and shaft assembly can also be refined to provide suitable rotational motion for different material quantities while maintaining reliable mechanical operation.

The vertical composting chamber and conical hopper can be further improved to support convenient material movement and simplify cleaning. Future modifications may include optimizing the hopper angle, outlet dimensions, and chamber geometry to reduce material accumulation and improve the transfer of processed material. The structural frame can also be redesigned to improve accessibility for inspection, maintenance, and component replacement.

Another area for future development is the improvement of the heating and conditioning section. The heating plate arrangement can be evaluated under different temperature and operating-duration conditions to identify suitable parameters for reducing excess moisture. The tray design can be modified to promote more uniform distribution of material during conditioning. Future experimental studies can measure initial and final. Further studies can investigate the suitability of the system for different types of biodegradable material, including vegetable residues, fruit peels, and other appropriate organic waste.

CHAPTER-14

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